

VIDYA SINGH

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CAREER OBJECTIVE

Computer Science and Artificial Intelligence student with hands-on experience in Python, C++, JavaScript, React.js, FastAPI, SQL, REST APIs, computer vision, machine learning, and backend development. Seeking a Software Engineer Intern role where I can build reliable applications, solve technical problems, integrate frontend and backend workflows, and contribute to production-ready software solutions.

EDUCATION

Meghnad Saha Institute of Technology (MSIT), Kolkata, West Bengal

Bachelor's Degree in Computer Science and Artificial Intelligence | Expected May 2026

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Fundamentals, Probability and Statistics, Linear Algebra with Computational Applications, Computer Science Fundamentals

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, HTML, CSS

Web and Backend: React.js, FastAPI, SQL, REST APIs, API Integration, Backend Development, Frontend Integration

Software Engineering: Data Structures and Algorithms, OOP, Problem Solving, Debugging, Git, Validation, Testing

AI/ML and Computer Vision: Machine Learning, Deep Learning, Computer Vision, OpenCV, TensorFlow, PyTorch, CNNs, Transfer Learning, Image Processing

Emerging AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Generative AI

EXPERIENCE

Hari Chand Anand & Co. | AI Engineer Intern | Jan 2026 - Present

- Developed a Python and OpenCV spare part recognition application with modular backend logic, structured outputs, and metadata lookup.
- Implemented preprocessing, feature extraction, feature matching, validation checks, and scoring workflows to improve recognition quality.
- Prepared datasets and evaluation scripts to measure accuracy, debug false matches, tune thresholds, and improve reliability.
- Integrated image input, camera capture, result display, and reusable recognition functions for an end-to-end application workflow.

Coincent.ai | Software Engineer Intern | Jun 2023 - Jul 2023 | Online

- Completed an internship focused on Python programming, artificial intelligence fundamentals, machine learning, and software implementation.
- Built a landmark detection project using image processing, model training, evaluation, and dataset-based prediction workflows.
- Strengthened practical understanding of Python modules, debugging, ML pipelines, classification, and model performance improvement.

PROJECTS

Spare Part Recognition System | Python, OpenCV, FastAPI-ready Logic

- Built a computer vision application with reusable backend functions, structured prediction results, and frontend GUI integration.
- Designed dataset indexing, metadata mapping, validation logic, CSV report generation, and evaluation workflows for accuracy tracking.
- Optimized preprocessing and matching logic to support reliable identification, debugging, and future REST API deployment.

Landmark Detection | Python, TensorFlow

- Developed a landmark classification model using CNN transfer learning with ResNet and InceptionV3 architectures.
- Improved accuracy from approximately 70% to 85-90% using data augmentation, hyperparameter tuning, and model optimization.

Multi-Modal Deepfake Detection System | Python, PyTorch, Deep Learning

- Developed a multi-modal detection pipeline using rPPG signals, visual facial features, audio spectrograms, and classification workflows.
- Implemented modular feature extraction, model evaluation, and performance tracking across video and audio modalities.
- Achieved 93.57% detection accuracy during project evaluation.